Recitation 5

In this problem, use **tidyverse functions only** (you are allowed to use any helper function such as mean(), median(), as.factor(), as.numeric(), and so on. Tidyverse package extensions can be used as well).

`transaction_data_1.csv` and `transaction_data_2.csv` datasets represent customer transactions at an online retail store specializing in electronics, clothing, and household items. The store collects transaction data to analyze customer behavior, improve recommendations, and optimize inventory. Each transaction includes demographic details, order details, and behavioral data such as payment method and discount usage. The goal is to use this data to personalize marketing efforts and enhance user experience.

The datasets include 800 purchases made over a three-month period and include the following variables:

transaction_data_1

- `Transaction_ID` - Unique identifier for each transaction.
- `Customer_ID` - Unique identifier for each customer.
- `Age` - Age of the customer.
- `Income` - Annual income of the customer in USD.
- `Discount_Applied` - Whether a discount was applied (Yes/No).
- `Customer_Rating` - Customer rating of the product (1-5 scale).
- `Quantity.Product_Price` - Quantity of units purchased followed by the product price (price per unit).

transaction_data_2

- `Transaction_ID` - Unique identifier for each transaction.
- `Gender` - Gender of the customer (Male, Female, Other).
- `Category` - Product category (Electronics, Clothing, Household).
- `Payment_Method` - Payment method used (Credit Card, PayPal, Debit Card, Other).
- `Shipping_Cost` - Cost of shipping in USD.
- `Delivery_Time` - Number of days taken for delivery.

Question 1:

- Clean the `transaction_data_1` and `transaction_data_2` datasets and make them tidy, that is, identify and fix all the issues introduced into the datasets.
- Merge `transaction_data_1` and `transaction_data_2`: the unmatched rows in either input should not be included in the output. Name the output `transaction_data`.
- Add a new variable, `Total_Amount`, to `transaction_data`, which represents the total cost of each transaction.

Use transaction_data for the remaining problems.

Question 2:

- Compute the average `Total_Amount` per transaction for each product `Category` and `Payment_Method` combination. Which payment method is associated with higher spending in each category?
- Create age groups (`<25`, `25-40`, `41-60`, `>60`). Within each age group, calculate the percentage of customers who used `Credit Card` as the payment method. What does this say about payment preferences by age?
- Split customers into `income` quartiles (below 25th percentile, between 25th and 50th percentiles, between 50th and 75th percentile, and above 75th percentile). Within each quartile, calculate the percentage of transactions with a discount applied. Interpret whether discounts tend to attract higher- or lower-income customers.
- Identify all customers who made purchases in **at least 2 different categories**. Among these multi-category shoppers, compute the average `Total_Amount` per transaction and average `Customer_rating`.
- Determine how many unique customers made multiple transactions. Among customers who made more than one transaction, calculate the average number of transactions per customer.

Question 3:

- Create a scatter plot of `Income` vs `Total_Amount`. Add marginal histograms on x and y axes, color points by `Discount_Applied`, and add a title and axis labels.